BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: 4471 / 4472

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q. Paper : II (Objective Type)

سوالیہ پرچہ: II (معروضی طرز)

QQ. 1. 1222 — Physics — (Inter Part - II) — (Session 2018-20 to 2020-22) — Paper (II) — (Group I) — (Objective)

Q. Time Allowed: 20 minutes — Maximum Marks: 17 — PAPER CODE 4471

Q1. A rubber ball of radius 2 cm has a charge of 5 µc on its surface, which is uniformly distributed. The value of E at its Centre is

(A) (B)

(C) (D)

Q2. The minimum value of charge on free particle is

(A) (B)

(C) (D)

Q3. During danger the 'eel' turn itself into a living battery. Then the potential difference between its head and tail can be upto

(A) (B)

(C) (D)

Q4. The sum of electric and magnetic force is called

(A) (B)

(C) (D)

Q5. Output waveform of sweep or time base generator is

(A) (B)

(C) (D)

Q6. Emf is induced due to change in

(A) (B)

(C) (D)

Q7. When the motor is just started, its back emf is

(A) (B)

(C) (D)

Q8. An A.C Voltmeter reads 220V, its peak value will be

(A) (B)

(C) (D)

Q9. When we acclerate the charge, which type of waves are produced?

(A) (B)

(C) (D)

Q10. A device used to detect very weak magnetic fields produced by brain is named as

(A) (B)

(C) (D)

Q11. The magnitude of voltage gain of an amplifier having r_{ie}=1 ohm, β = 100 and R_c=200 ohm is

(A) (B)

(C) (D)

Q12. Which one is used as temperature sensor in electrical circuit?

(A) (B)

(C) (D)

Q13. The rest mass of photon is

(A) (B)

(C) (D)

Q14. The materialization of energy takes place in the process of

(A) (B)

(C) (D)

Q15. The unit of Rydberg's constant R_h is

(A) (B)

(C) (D)

Q16. The unit of decay constant is

(A) (B)

(C) (D)

Q17. Half life of radioactive isotope of Iodine-131 is

(A) (B)

(C) (D)

Board typos kept verbatim: 'acclerate', 'photoelectric' lower-case in option, 'E at its Centre', '5 µc'.

Paper : II (Essay Type)

پرچہ: II (انشائیہ طرز)

QSection I. 1222 — Physics — (Inter Part - II) — (Session 2018-20 to 2020-22) — Paper (II) — (Group I) — (Subjective)

Q. Time Allowed: 2.40 hours — Maximum Marks: 68

Q. Section ---------- II — (8 × 3 = 24)

Q. Note: Attempt any three questions.

GROUP – II

Q. Paper : II (Objective Type)

سوالیہ پرچہ: II (معروضی طرز)

QQ. 1. 1222 — Physics — (Inter Part - II) — (Session 2018-20 to 2020-22) — Paper (II) — (Group II) — (Objective)

Q. Time Allowed: 20 minutes — Maximum Marks: 17 — PAPER CODE 4472

Q1. Photo copier and injek printer are the application of :

(A) (B)

(C) (D)

Q2. SI unit of electric flux is:

(A) (B)

(C) (D)

Q3. When the internal resistance of source is equal to the load maximum power dissipated is

(A) (B)

(C) (D)

Q4. Unit of magnetic flux density is

(A) (B)

(C) (D)

Q5. When a charge is projected perpendicular to uniform magnetic field its path is:

(A) (B)

(C) (D)

Q6. If the angular frequency of A.C Generator increased to double, the time period would become

(A) (B)

(C) (D)

Q7. "Eddy current" are set up in a direction:

(A) (B)

(C) (D)

Q8. When effective value of current is 10. What is its peak value?

(A) (B)

(C) (D)

Q9. Which are the Substance called _____ which undergo plastic deformation until they break.

(A) (B)

(C) (D)

Q10. Choke consumes extremly small.

(A) (B)

(C) (D)

Q11. The size of base in a transistor is

(A) (B)

(C) (D)

Q12. _____ is the building block of every complex electronic circuit.

(A) (B)

(C) (D)

Q13. The unit of work function is

(A) (B)

(C) (D)

Q14. Compton's Shift will be maximum at the angle of

(A) (B)

(C) (D)

Q15. Production of X-rays is the reverse process of

(A) (B)

(C) (D)

Q16. The Binding energy for Helium is given by

(A) (B)

(C) (D)

Q17. After two half-lives the number of decayed nuclei of an element are

(A) (B)

(C) (D)

Board typos kept verbatim: 'injek printer', 'Electro magnetism', 'extremly small', 'perpendicular to the flux' (option D split across lines).

Paper : II (Essay Type)

پرچہ: II (انشائیہ طرز)

QSection I. 1222 — Physics — (Inter Part - II) — (Session 2018-20 to 2020-22) — Paper (II) — (Group II) — (Subjective)

Q. Time Allowed: 2.40 hours — Maximum Marks: 68

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Q. Note: Attempt any three questions.